Day 2

TCP/IP Suite Theory (1 hour):What is the TCP/IP Suite? Layers of TCP/IP: Application, Transport, Internet, and Network Access. Practical (1 hour):Analyse packet headers to identify the TCP/IP layer using Wireshark. Task: Compare OSI and TCP/IP models with real-world examples.

1. **Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.**

ANS :

TCP/IP Layer	Protocol Found	Example Packet Purpose
Application	DNS / HTTPS	Resolves flipkart.com and loads webpage data
Transport	TCP	Establishes reliable connection
Internet	IP (IPv4)	Routes packets between devices
Network Access	ARP / Ethernet II	Transfers frames within local network

No.	Time	Source	Destination	Protocol	Lengtl	Info
7153	22.654788900	23.212.0.20	192.168.0.108	TCP	1466	443 → 58645 [ACK] Seq=809527 Ack=5938 Win=64128 Len=1412 [TCP PDU reassembled in 7155]
7154	22.654791600	23.212.0.20	192.168.0.108	TCP	1466	443 → 58645 [ACK] Seq=810939 Ack=5938 Win=64128 Len=1412 [TCP PDU reassembled in 7155]
7155	22.654794000	23.212.0.20	192.168.0.108	TLSv1.3	1357	Application Data
7156	22.654796400	23.212.0.20	192.168.0.108	TLSv1.3	1466	Application Data
7157	22.654799700	23.212.0.20	192.168.0.108	TCP	1466	443 → 58645 [ACK] Seq=815066 Ack=5938 Win=64128 Len=1412 [TCP PDU reassembled in 7165]
7158	22.654801700	23.212.0.20	192.168.0.108	TCP	1466	443 → 58645 [ACK] Seq=816478 Ack=5938 Win=64128 Len=1412 [TCP PDU reassembled in 7165]
7159	22.654804000	23.212.0.20	192.168.0.108	TCP	1466	443 → 58645 [ACK] Seq=817890 Ack=5938 Win=64128 Len=1412 [TCP PDU reassembled in 7165]
7160	22.654806900	23.212.0.20	192.168.0.108	TCP	1466	443 → 58645 [ACK] Seq=819302 Ack=5938 Win=64128 Len=1412 [TCP PDU reassembled in 7165]
7161	22.654874700	23.212.0.20	192.168.0.108	TCP	1466	443 → 58645 [ACK] Seq=820714 Ack=5938 Win=64128 Len=1412 [TCP PDU reassembled in 7165]
7162	22.654877500	23.212.0.20	192.168.0.108	TCP	1466	443 → 58645 [ACK] Seq=822126 Ack=5938 Win=64128 Len=1412 [TCP PDU reassembled in 7165]
7163	22.654888600	23.212.0.20	192.168.0.108	TCP	1466	443 → 58645 [ACK] Seq=823538 Ack=5938 Win=64128 Len=1412 [TCP PDU reassembled in 7165]
7164	22.654890800	23.212.0.20	192.168.0.108	TCP	1466	443 → 58645 [ACK] Seq=824950 Ack=5938 Win=64128 Len=1412 [TCP PDU reassembled in 7165]
7165	22.654892000	23.212.0.20	192.168.0.108	TLSv1.3	326	Application Data
7166	22.655035100	192.168.0.108	23.212.0.20	TCP	54	58645 → 443 [ACK] Seq=5973 Ack=826634 Win=524032 Len=0
7167	22.706130000	23.212.0.20	192.168.0.108	TCP	60	443 → 58645 [ACK] Seq=826634 Ack=5973 Win=64128 Len=0
7168	23.138489300	TPLink_88:0f:c0	Broadcast	ARP	42	Who has 192.168.0.106? Tell 192.168.0.1
7169	23.138585400	TPLink_88:0f:c0	Broadcast	ARP	42	Who has 192.168.0.107? Tell 192.168.0.1

2. **Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).**

ANS :

OSI Model	TCP/IP Model	Real-World Example
Application	Application	Sending WhatsApp message
Presentation	Application	Instagram image formatting and encryption
Session	Application	WhatsApp voice call session
Transport	Transport	TCP connection while ordering food on Zomato
Network	Internet	IP routing when accessing YouTube
Data Link	Network Access	Wi-Fi MAC address communication
Physical	Network Access	Wi-Fi signals or Ethernet cable

- **Key Difference**

- OSI Model has **7 layers**

- TCP/IP Model has **4 layers**

- TCP/IP combines OSI's Application, Presentation, and Session layers into one Application layer.

- TCP/IP combines OSI's Data Link and Physical layers into one Network Access layer.

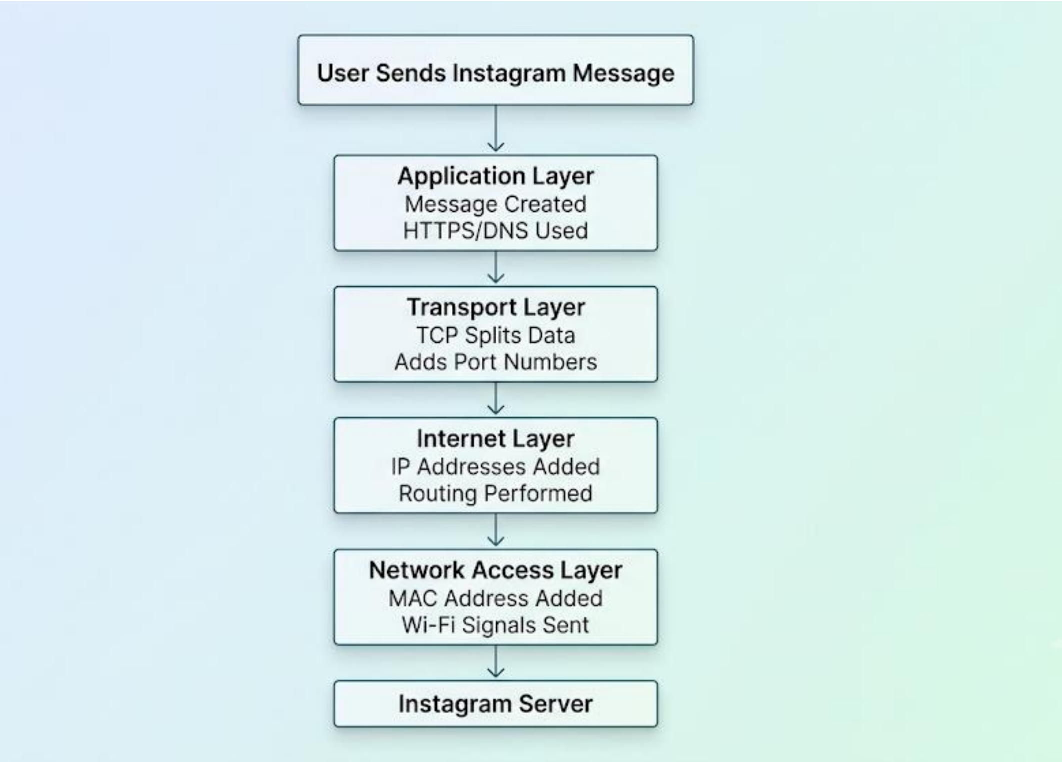
3. Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines. Hint : Look for fields like Source Port, Destination Port, Sequence Number, etc.

ANS :

Field	**Function**
Source Port	Identifies sender application
Destination Port	Identifies receiver application
Sequence Number	Tracks packet order
Acknowledgment Number	Confirms received packets
Header Length	Specifies header size
Flags	Controls connection (SYN, ACK, FIN)
Window Size	Controls data flow amount
Urgent Pointer	Indicates urgent data

4. **Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).**

ANS :



- **Layer Functions**

  - Application Layer → Creates and encrypts message.
  - Transport Layer → Splits data into packets.
  - Internet Layer → Adds IP addresses and routes packets.
  - Network Access Layer → Sends packets over Wi-Fi/Ethernet.

5. Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

ANS :

- **Packet Chosen: TCP Packet**
- **Protocol:** TCP
- **TCP/IP Layer:** Transport Layer
- **How I Identified It:**

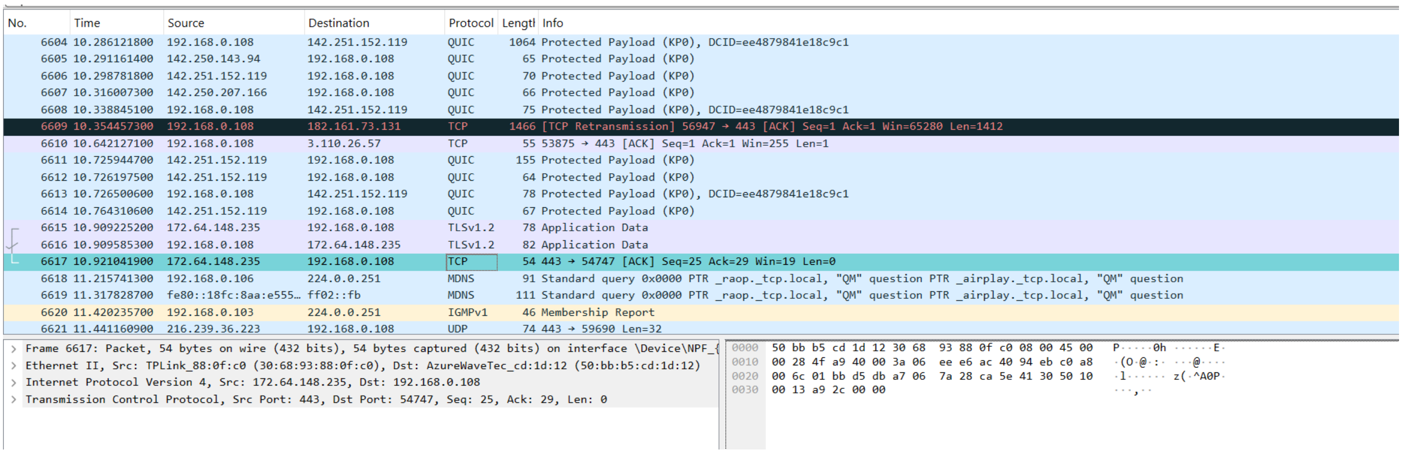
  - The Protocol column showed "TCP".

  - Packet details contained Source Port, Destination Port, Sequence Number, and Acknowledgment Number fields.

- **Real-World Action Represented:**

  - This packet represents the reliable transfer of data when sending a WhatsApp message, loading Instagram posts, or booking movie tickets on BookMyShow.

  - TCP ensures that all data reaches the destination correctly and in order.



No.	Time	Source	Destination	Protocol	Lengtl	Info
6604	10.286121800	192.168.0.108	142.251.152.119	QUIC	1064	Protected Payload (KP0), DCID=ee4879841e18c9c1
6605	10.291161400	142.250.143.94	192.168.0.108	QUIC	65	Protected Payload (KP0)
6606	10.298781800	142.251.152.119	192.168.0.108	QUIC	70	Protected Payload (KP0)
6607	10.316007300	142.250.207.166	192.168.0.108	QUIC	66	Protected Payload (KP0)
6608	10.338845100	192.168.0.108	142.251.152.119	QUIC	75	Protected Payload (KP0), DCID=ee4879841e18c9c1
6609	10.354457300	192.168.0.108	182.161.73.131	TCP	1466	[TCP Retransmission] 56947 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1412
6610	10.642127100	192.168.0.108	3.110.26.57	TCP	55	53875 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
6611	10.725944700	142.251.152.119	192.168.0.108	QUIC	155	Protected Payload (KP0)
6612	10.726197500	142.251.152.119	192.168.0.108	QUIC	64	Protected Payload (KP0)
6613	10.726500600	192.168.0.108	142.251.152.119	QUIC	78	Protected Payload (KP0), DCID=ee4879841e18c9c1
6614	10.764310600	142.251.152.119	192.168.0.108	QUIC	67	Protected Payload (KP0)
6615	10.909225200	172.64.148.235	192.168.0.108	TLSv1.2	78	Application Data
6616	10.909585300	192.168.0.108	172.64.148.235	TLSv1.2	82	Application Data
6617	10.921041900	172.64.148.235	192.168.0.108	TCP	54	443 → 54747 [ACK] Seq=25 Ack=29 Win=19 Len=0
6618	11.215741300	192.168.0.106	224.0.0.251	MDNS	91	Standard query 0x0000 PTR _raop._tcp.local, "QM" question PTR _airplay._tcp.local, "QM" question
6619	11.317828700	fe80::18fc:8aa:e555…	ff02::fb	MDNS	111	Standard query 0x0000 PTR _raop._tcp.local, "QM" question PTR _airplay._tcp.local, "QM" question
6620	11.420235700	192.168.0.103	224.0.0.251	IGMPv1	46	Membership Report
6621	11.441160900	216.239.36.223	192.168.0.108	UDP	74	443 → 59690 Len=32

> Frame 6617: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{
> Ethernet II, Src: TPLink_88:0f:c0 (30:68:93:88:0f:c0), Dst: AzureWaveTec_cd:1d:12 (50:bb:b5:cd:1d:12)
> Internet Protocol Version 4, Src: 172.64.148.235, Dst: 192.168.0.108
> Transmission Control Protocol, Src Port: 443, Dst Port: 54747, Seq: 25, Ack: 29, Len: 0

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0000  50 bb b5 cd 1d 12 30 68  93 88 0f c0 08 00 45 00   P·····0h ······E·
0010  00 28 4f a9 40 00 3a 06  ee e6 ac 40 94 eb c0 a8   ·(O·@·:· ···@····
0020  00 6c 01 bb d5 db a7 06  7a 28 ca 5e 41 30 50 10   ·l······ z(·^A0P·
0030  00 13 a9 2c 00 00                                  ···,··
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